1 Introduction

This lab report discusses the experiments conducted on various op-amp circuits. The main aim of this lab was to analyze the input-output relationships and gain of op-amp circuits. The experiments involved calculating the necessary values for the components, constructing the circuits on CircuitLab, and observing the output responses of the circuits. The lab experiments included tasks such as finding the input-output relationship and gain, deriving the expression for input-output, proving that the amplifier works as a summing amplifier, finding the gain of the circuit in different switch positions, and finding the output of various op-amp circuits.

1.1 Preliminary Work

Before the Lab Tasks, we need to calculate tasks answers before we construct the circuits on CircuitLab.

1. For task 1 we are asked to find the input-output relationship, and the gain of the circuit if Rf is 10k and R1 is 1k on the circuit shown with ideal op-amp properties on Figure 1.

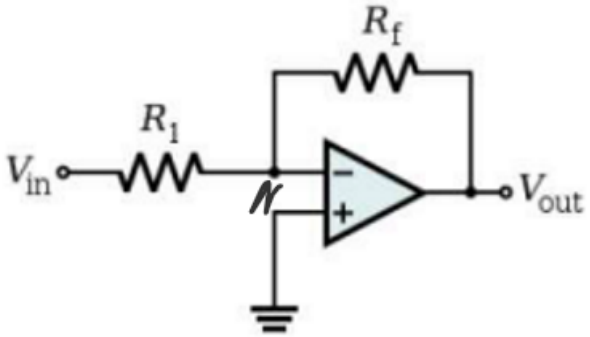


Figure 1: Circuit to use in task 1

(a) Firstly, we have virtual short circuit formula as $V_p = V_n$ and i_{in} and i_p is equal to zero

(b) Then we will apply KCL to node N.
$(V_p - V_{in})/R_1 + (V_p - V_{out})/R_f + i_{in} = 0$,
since $V_p = V_n$ and $V_n = 0, V_p = 0$,
$V_{in}/R_1 + V_{out}/R_f = 0$,
$V_{out} = (-R_f/R_1)V_{in}$

(c) If $R_f = 10k$ and $R1is1k$
$V_{out} = -10V_{in}$

2. For task 2 we are asked to derive the expression for input-output. What value of feedback resistor Rf is needed to give an amplification equal to 2 when R1=10k on Figure 2.

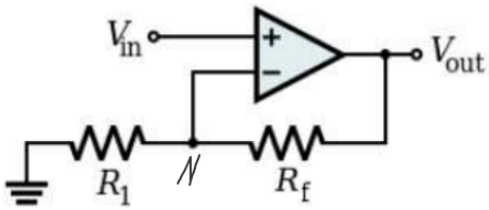


Figure 2: Circuit to use in task 2

(a) Firstly, we have virtual short circuit formula as $V_p = V_n$
since $V_p = V_n$ is leads us to $V_n = V_{in}$

(b) Then we will apply KCL to node N.

$(V_{in} - 0V)/R_1 + (V_{in} - V_{out})/R_f + i_{in} = 0$, i_{in} and i_p is equal to zero
$V_{in}/R_1 + (V_{in} - V_{out})/R_f = 0$,
$(1 + R_f/R_1)V_{in} = V_{out}$,

(c) $(1 + R_f)/R_1 = 2$ from that equation we can obtain R_f as 10k

3. For task 3 we are asked to prove that the amplifier shown in Figure 3 works as a summing amplifier.

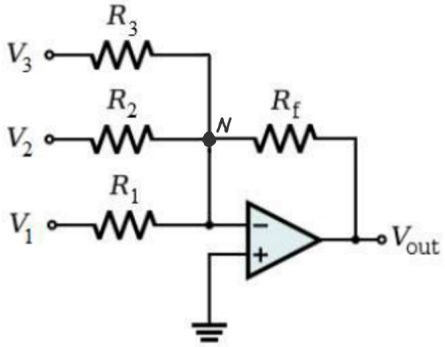


Figure 3: Circuit to use in task 3

(a) $V_p = 0$ since $V_n = V_p$ and $V_n = 0$

(b) We will apply KCL for node N.
$(0 - V_1)/R_1 + (0 - V_2)/R_2 + (0 - V_3)/R_3 + (0 - V_{out})/R_f = 0$
$V_{out} = -[(R_f/R_1)V_1 + (R_f/R_2)V_2 + (R_f/R_3)V_3]$

(c) If $R_1 = R_2 = R_3 = R_s$
then

(d) $V_1 = (-R_F/R_s)(V_1 + V_2 + V_3)$
This is linear sum of input voltages.

4. For task 4 we are asked to find the gain of the following circuit when the switch is in A and B positions, separately.

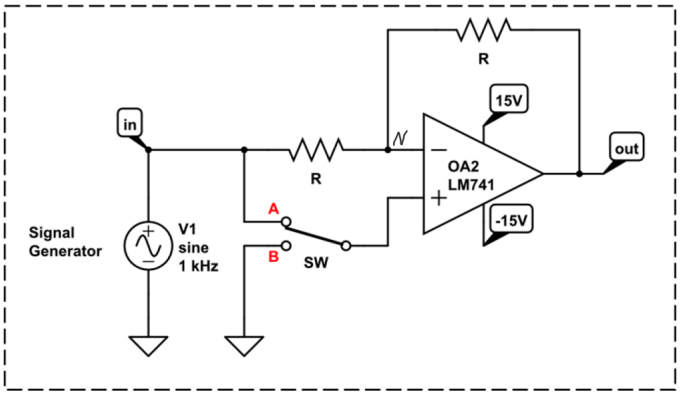


Figure 4: Circuit to use in task 4

(a) We will apply first KCL for node N.
$(V_n - V_{in})/R + (V_n - V_{out})/R + i_{in} = 0$ and i_{in} and i_p is equal to zero

(b) $V_p = V_{in}$ and $V_n = V_{in}$, because $V_p = V_n$
$(V_{in} - V_{in})/R + (V_{in} - V_{out})/R = 0$

(c) $(V_{in} - V_{out})/R = 0$ with that equation we will obtain $V_{in} = V_{out}$

5. If we turn switch to B, we will have Figure 5.

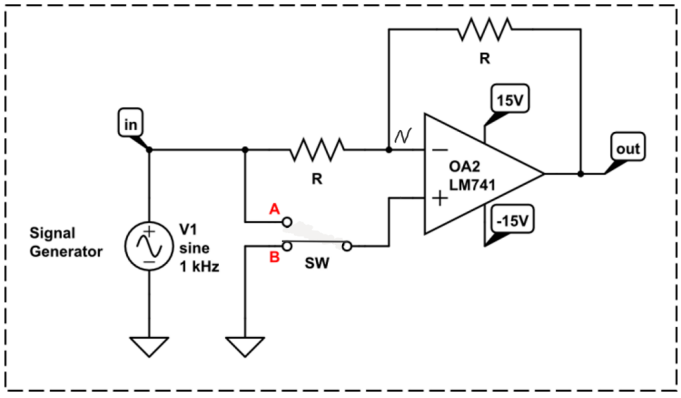


Figure 5: Switch turned to B in task 4.

(a) Firstly, we have virtual short circuit formula as $V_p = V_n$
$V_p = 0V$ then $V_n = 0V$

(b) Then we will apply KCL to node N.
$(V_n - V_{in})/R + (V_n - V_{out})/R + i_n = 0$, i_n and V_n is equal to 0.

(c) $V_{in}/R + V_{out}/R = 0$ equals to $(V_{in} + V_{out})/R = 0$. That equations will leads us to $V_{out} = -V_{in}$

6. We are asked to find the output of circuits shown below (a, b, c, d) and all OpAmps are assumed ideal.

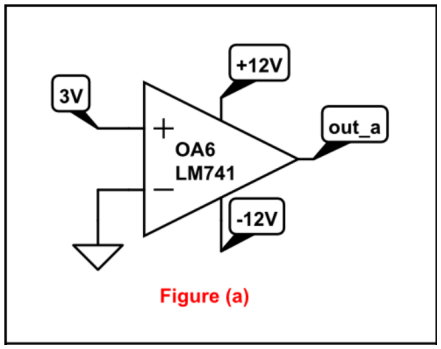


Figure 6: circuit a, task a

(a) In the circuit a (Figure 6), non inverting type and power supply is +12, -12 Volts the output of OpAmp is difference between +V side port voltage and -V side port voltage.
$+V = 3V$
$-V = 0V$
$V_{out_a} = 3V + 0V$ Which is equal to $V_{out_a} = 3V$.

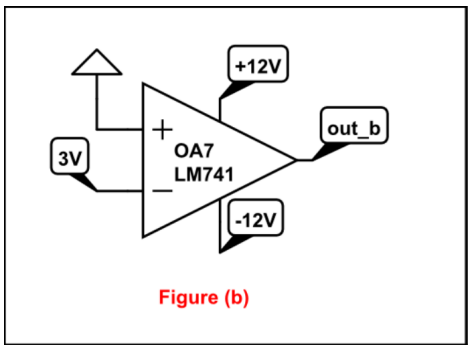


Figure 7: circuit b, task b

(b) In the circuit b (Figure 7), inverting type and power supply is +12, -12 Volts the output of OpAmp is difference between +V side port voltage and -V side port voltage.
$+V = 0V$
$-V = 3V$
$V_{out_a} = 0V - 3V$ Which is equal to $V_{out_b} = -3V$.

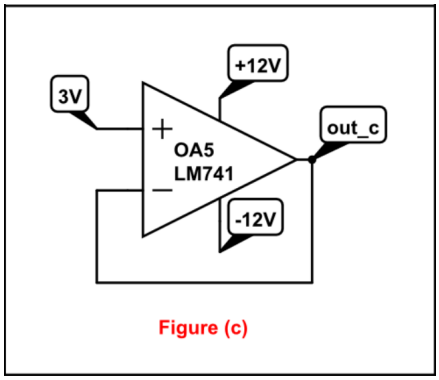


Figure 8: circuit c, task c

(c) In the circuit c (Figure 8), this is the case of voltage follower OpAmp.
$+V = -V = 3V$
$V_{out_c} = -V$
$V_{out_c} = 3V$ the output voltage is 3V.

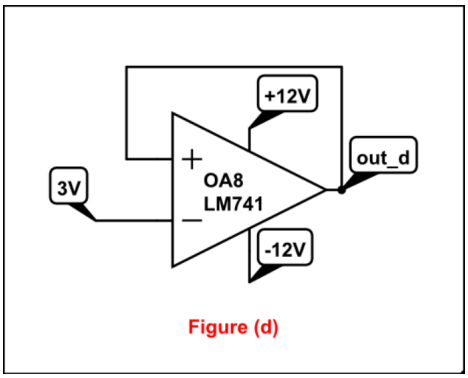


Figure 9: circuit d, task d

(d) The output of U3 is connected to V_{out_d}, which means that the output voltage of U3 is the same as the input voltage.
$V_{out_d} = V_{in} = 6V$

1.2 Lab Tasks

We have completed preliminary work with calculations, now we will check our calculations on CircuitLab simulations for task 1, 2, 3, 4 and a, b, c.

1.2.1 Task 1.

1. First of all, we have constructed the circuit (Figure 10) on the circuit lab.

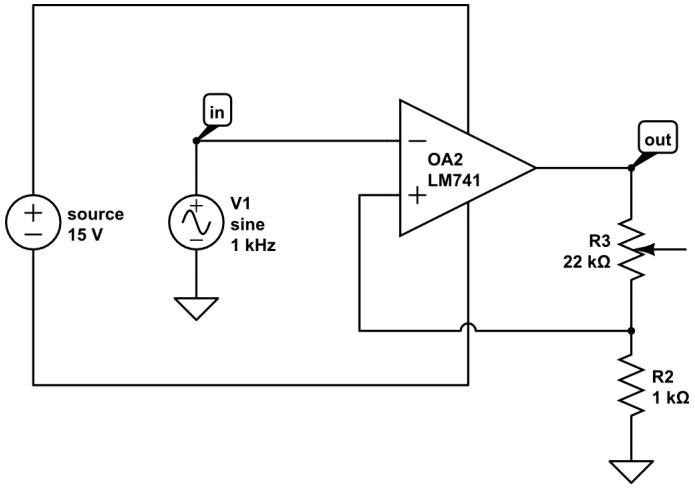


Figure 10: The circuit we have designed on circuit lab.

2. With Sweeper Parameters, 1k, 2k, 5k. We obtained that graphic.

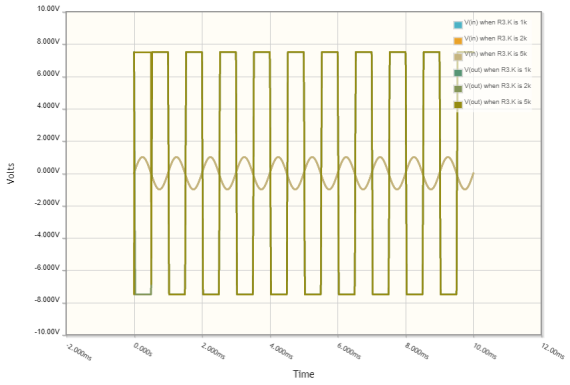


Figure 11: The time domain plot of the circuit (Figure 10) we have designed on circuit lab.

3. Volts are shown like in the Figure 11,
4. Now we will adjust the resistance of R3, and use sweeper in circuit lab with 1k, 2k, and 5k.
5. Then we will find resistance value for R3 to find clippings.

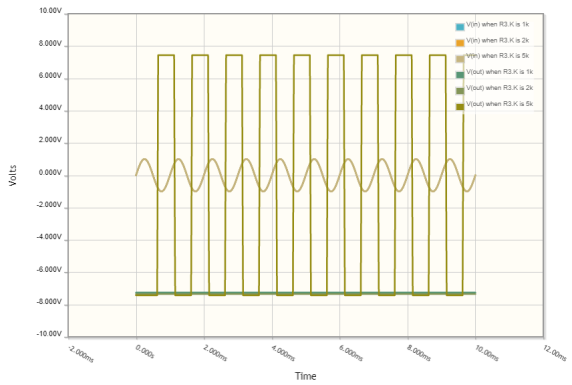


Figure 12: The time domain plot of the circuit (Figure 10) we have designed on circuit lab. Only 5k clips on 2ohm for R3.

6. When we adjust the R3 resistance to 2ohm, we see only 5k clips in this situation.

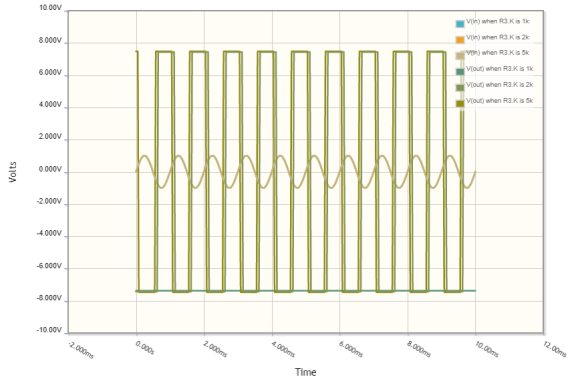


Figure 13: The time domain plot of the circuit (Figure 10) we have designed on circuit lab. 5k and 2k both clips on 5 ohm for R3.

7. When we adjust the R3 resistance to 5ohm, we see 5k and 2k both clips in this situation.

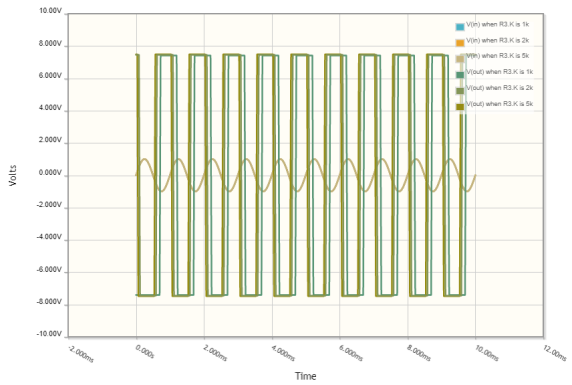


Figure 14: The time domain plot of the circuit (Figure 10) we have designed on circuit lab. All of the k values (1,2,5) clips on 7ohm for R3.

8. When we adjust the R3 resistance to 7ohm, we see all of the k values clips in this situation.

9. now we will change sweep type from custom to linear, system will generate response for 1k to 10k with every change with 1k in resistor.

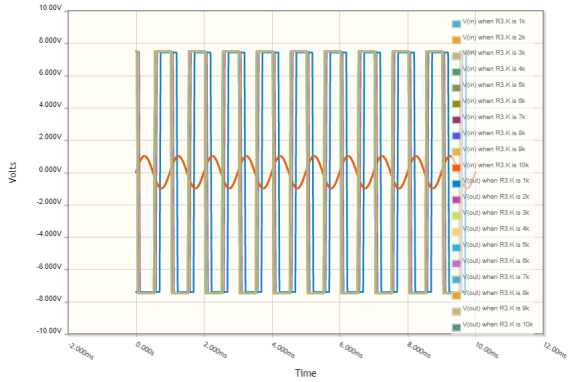


Figure 15: The time domain plot of the circuit (Figure 10) we have designed on circuit lab. Plot sweeper type changed to linear to 1k to 10k with steps 1k.

10. When we export the plot all of the V(out) values are starts clipping between 1k and 9k but not on the 10k value.

1.2.2 Task 2.

In that task 2. We will do same steps with Task 1 but on Inverting Amplifier.

1. We have constructed the circuit on the Circuit Lab simulation.

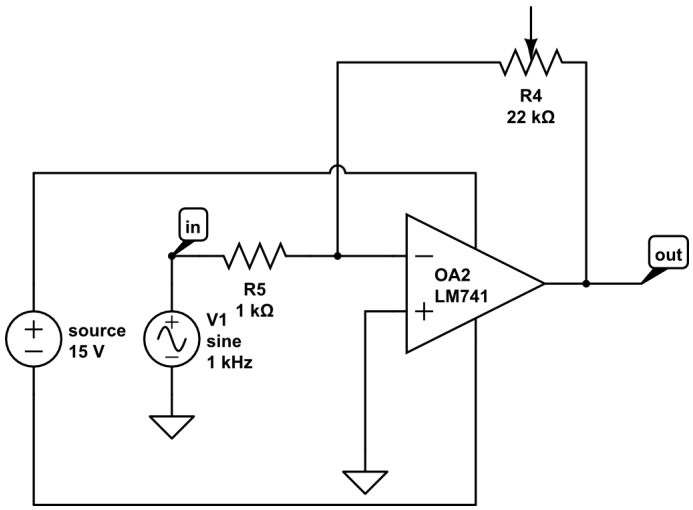


Figure 16: The circuit for task 2. Inverting Amplifier.

2. We are exporting the plot when we run the system in in time domain with sweep parameter with linear type to plot for resistance 1k to 10k with every 1k step.

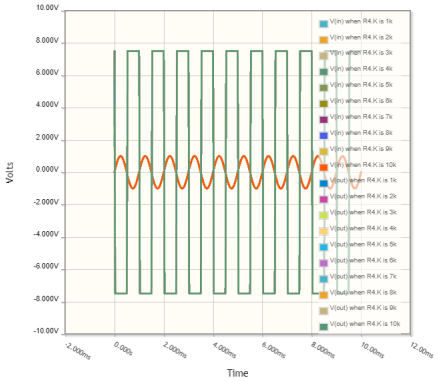


Figure 17: Plot of figure 16.

3. we see the same result as we get on the 1.2.1.10, values are starts clipping between 1k and 9k but not on the 10k value.

1.2.3 Task 3.

On the task 3, we will control if our calculations are correct when we tried to solve task 4 in preliminary work.

1. We have constructed Figure 4 on the circuit lab

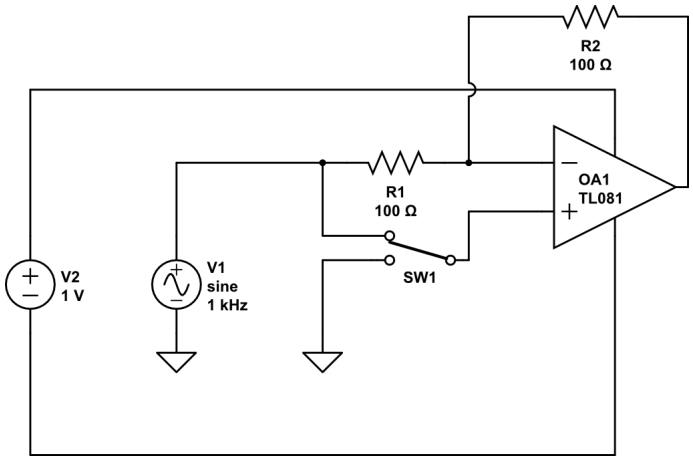


Figure 18: Task 3 on circuit lab.

2. now if we calculate when the switch on the top (a). We will have these calculations from the simulation.

3. We have constructed Figure 4 on the circuit lab

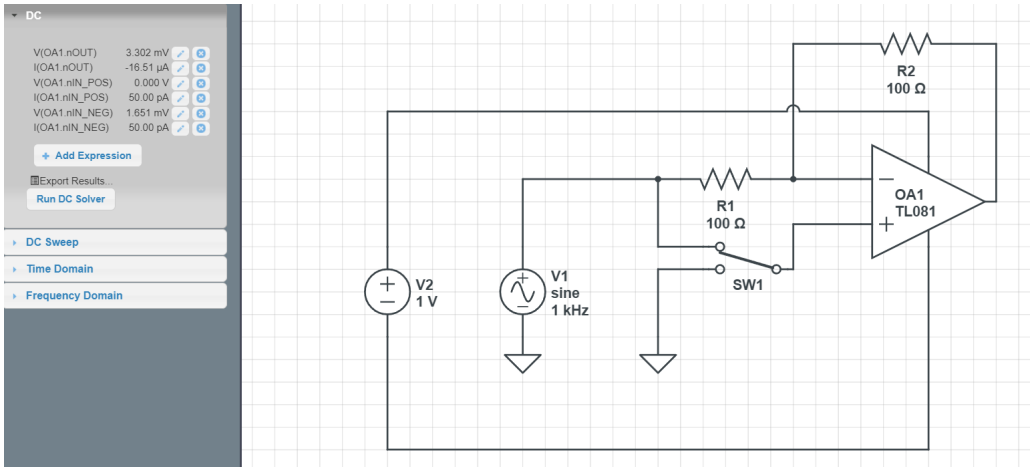


Figure 19: Values of task 3.

4. As we can see from figure 19 our calculations are near to simulation, we have found 3V and simulation shows 3.3 volts.

5. if we turn the switch bottom position we will have,

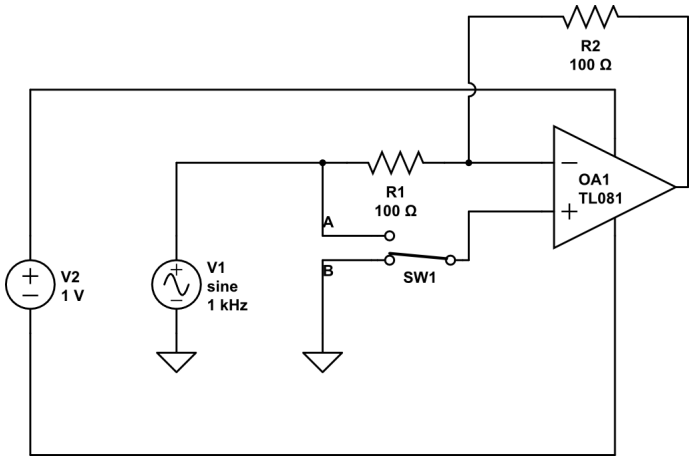


Figure 20: The switch on Figure 18 turned to bottom statement (B).

6. In that statement simulation will give us the values as shown below.

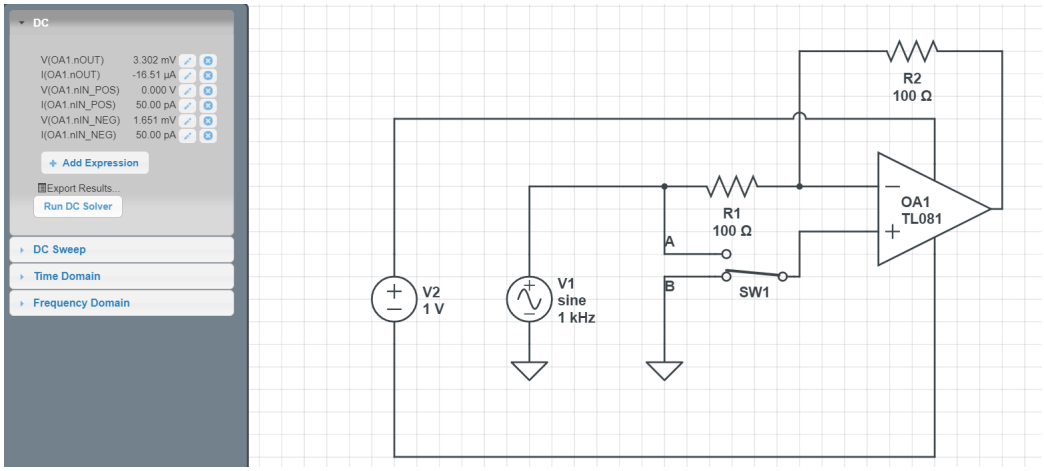


Figure 21: Values of task 3 while switch turned to bottom state.

7. As we can see our calculations are near to simulation, we have found 3V and simulation shows 3.3 volts.

1.2.4 Task 4.

On the last task we will check if our calculations on the task a,b,c,d (1.1.6) are correct.

1. first we have constructed the circuit shown in figure 6 on circuit lab.

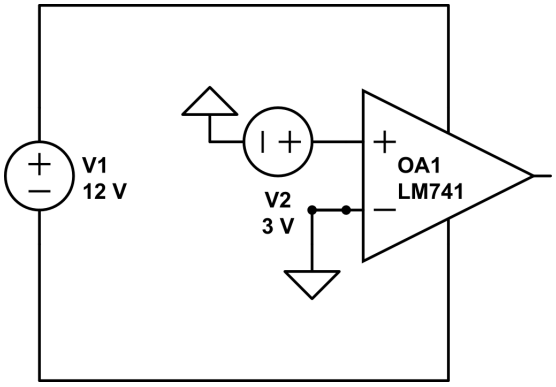


Figure 22: Task 4 a. on circuit lab.

2. as we run the system on the time domain we can see our calculations were 3V but simulation gives 6V as shown below on figure 23.

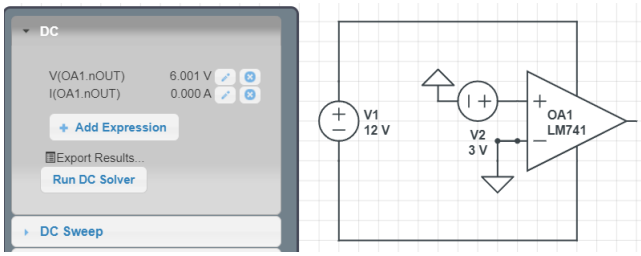


Figure 23: Task 4 a. on circuit lab. With values of output.

3. for the figure 7, we will apply same steps on the 1.2.4.1.

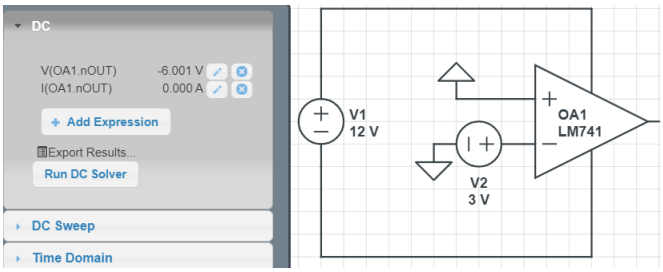


Figure 24: Task 4 b. on circuit lab.

4. as we run the system on the time domain we can see our calculations were -3V but simulation gives -6V.

5. for the figure 8, we will apply same steps on the 1.2.4.1.

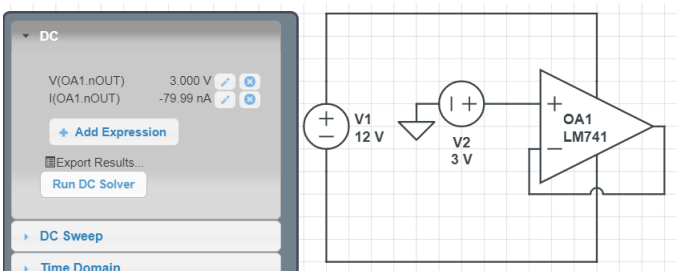


Figure 25: Task 4 c. on circuit lab. With values of output.

6. our calculation with the simulation is same 3V.
7. for the figure 9, we will apply same steps on the 1.2.4.1.

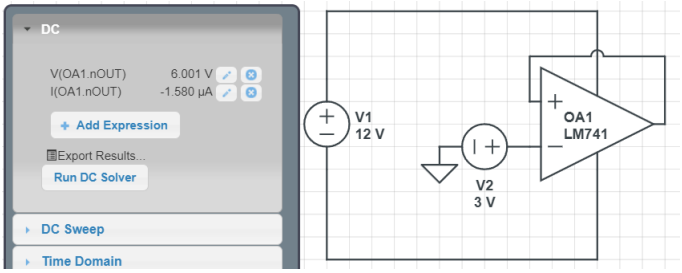


Figure 26: Task 4 d. on circuit lab. With values of output.

8. our calculation with the simulation is same 6V.

2 Conclusion

This lab report successfully analyzed op-amp circuits by performing various tasks, such as calculating component values, building circuits on CircuitLab, and observing circuit outputs. The experiments taught us about input-output relationships and the gain of op-amp circuits. We learned that component values are crucial in determining the output response of the circuits. However, we did encounter some discrepancies in our calculations, specifically in finding different values for 1.2.4.2 and 1.2.4.4, indicating an error in our calculations. Despite this setback, we still gained valuable knowledge from the experiments and made accurate observations of circuit responses. Overall, this lab report has given us valuable insights into op-amp circuits, while also highlighting the importance of being careful and accurate in our calculations.

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